

Linux LVM Lab

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Purpose: The purpose of this lab is to practice configuring LVM which is a software-defined storage feature on Linux systems. LVM allows you to pool multiple storage devices into one virtual drive by abstracting the physical devices. By supporting this, it allows storage to be increased and decreased on demand to meet your needs. The abstraction also allows software to perform RAID, snapshots, and a variety of other features.

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4. Create and Manage Logical Volumes:
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6. Apply Mirroring to a LV:
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Section #1 Introduction and Setup:

1.1 What is LVM? What are the parts that make it up?

LVM (Logical Volume Manager) is a Linux software defined storage feature that allows users to manage variable storage needs on Linux devices. LVM features many features that make it a flexible storage management option on Linux systems.

Features	Explanation
1. Pool storage devices	LVM allows you to pool storage devices together to create larger storage sizes. This feature also makes managing multiple storage devices easier since multiple storage devices can be controlled as though they are a single device.
2. Dynamic resizing	LVM allows you to dynamically resize your storage. This means that you can resize the storage without having to shut down your system. If you have greater storage needs you can add the new storage device to LVM while the system is live. Removing storage devices and shrinking the size of LVM storage can also occur while the system is live.

3. Snapshots	LVM supports snapshots which are virtual images of an LVM drive. It supports incremental and full snapshots.
4. Striping	LVM supports striping which allows data to be written randomly between different drives. Striping is RAID 0 and offers no data redundancy or protection however. It is mainly used to increase read/write speeds. If one drive is lost the data is gone and cannot be recovered.
5. Mirroring	Mirroring is supported as well. Mirroring, or RAID 1, provides fault tolerance and data redundancy by mirroring data across two drives. The only downside is that mirroring makes you lose storage capacity. If you had two 50 GB drives in a RAID 1 array only 50 GB would be used for storage since the other 50 GB is used for mirroring.
6. Migrating Volumes	LVM migration allows data to be moved from one PV to another with no system downtime. If a PV is having issues and you want to be proactive you can use this ability to move the data to a healthy PV.

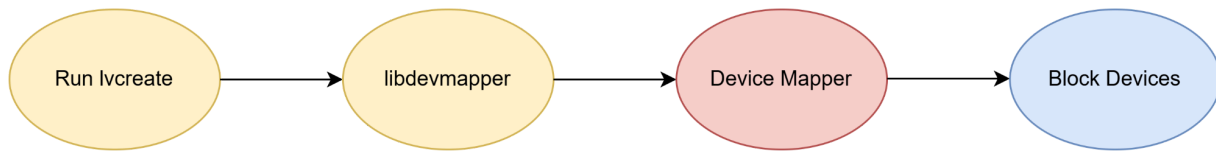
1.2 Why is LVM used?

LVM is mainly used in environments with changing storage requirements. A great example of a LVM application is using it for a database server. Database servers will have growing storage capacity needs as a business grows and acquires more data from customer transactions and other sources and having LVM would be a great feature for one. Since LVM will allow the database administrator to add more storage dynamically as the needs grow, it supports data redundancy via mirroring, and it allows for periodic snapshots to be taken not only would the administrator have the ability to meet the growing storage requirements they will also be able to ensure that the integrity and availability of the database data is secured.

LVM can also be used in non-enterprise use cases as well. You can use LVM in your home to manage storage for your Linux desktop system. It will make managing various storage devices a lot more streamlined since you won't have to manage each device on its own and only the volume group it is a part of. It can also help you ensure that your data is protected in the same manner as the database example. It wouldn't hurt to use mirroring to ensure that your data will be safe in the vent of a storage device failure.

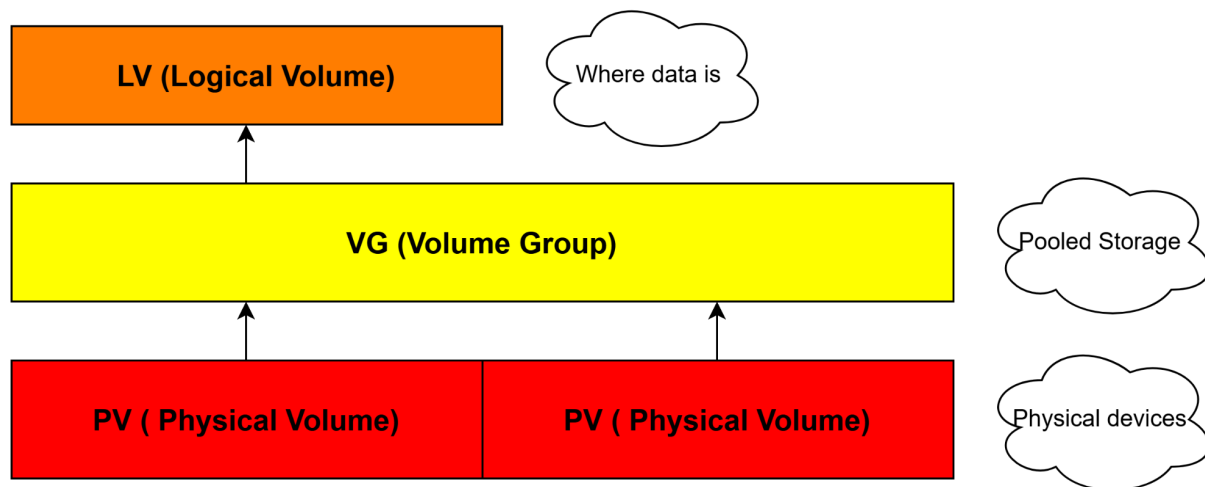
1.3 How does LVM work?

LVM is software defined storage and uses a mix of kernel space and user space programs to manage storage.



So for example when you run the `lvcreate` command to create a logical volume the command will run in user space but will make a call to the kernel space using `libdevmapper` library functions. LVM communicates with the Device Mapper kernel space feature that allows the Linux system to map physical drives into virtual drives. So when `lvcreate` is used the kernel will be able to map the virtual logical volume to physical sectors on a drive.

LVM relies on three levels of abstraction which in order are physical volumes, volume groups, and logical volumes.

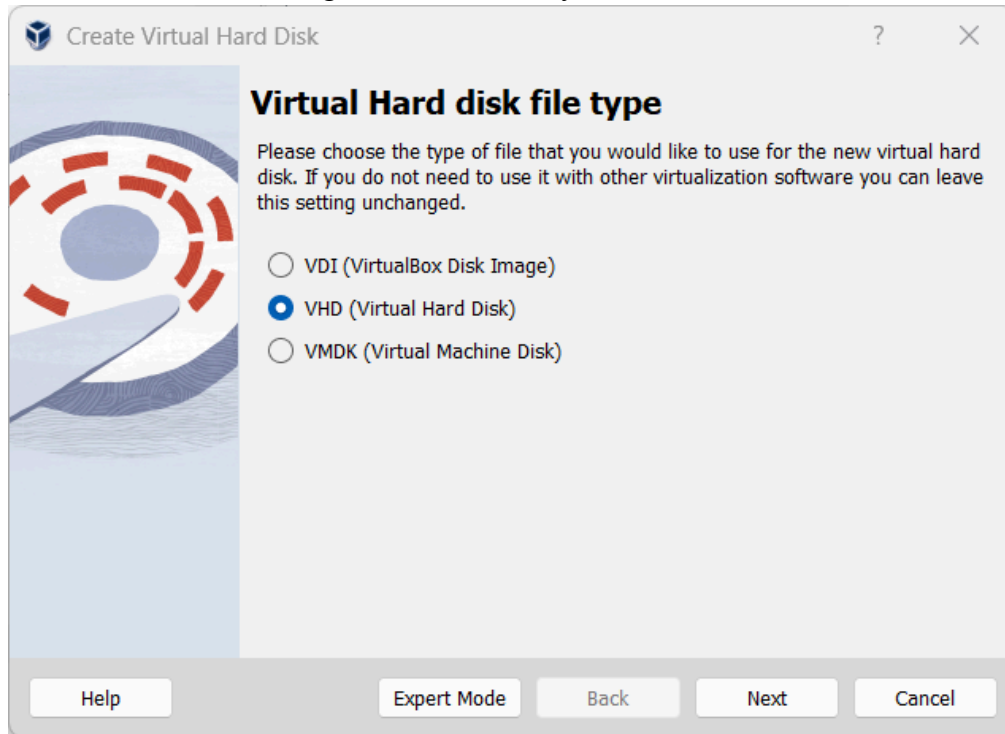


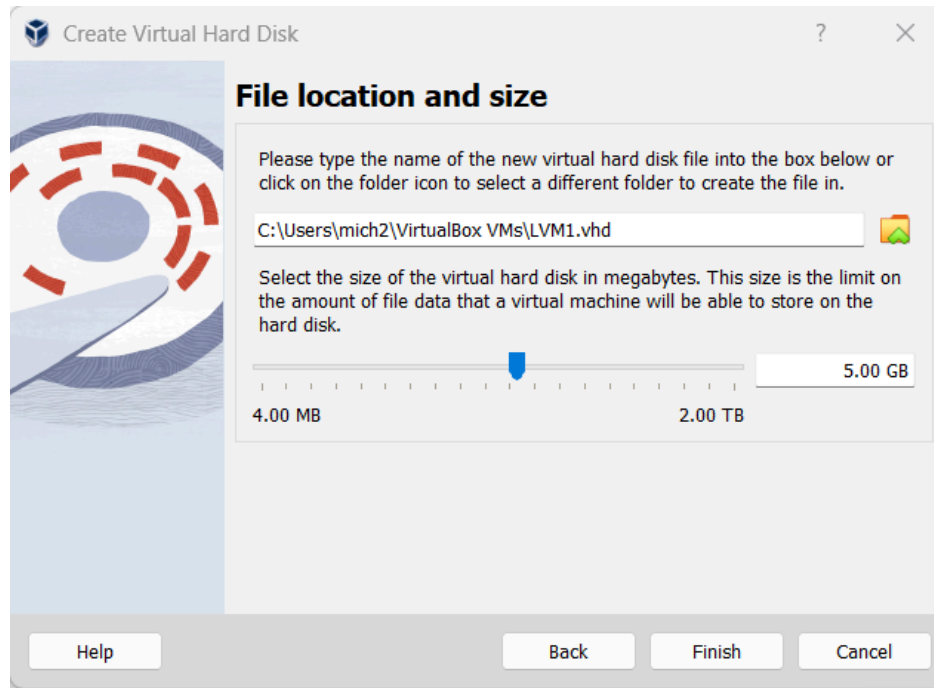
LVM abstracts physical drives via physical volumes. Physical volumes are essentially virtualized storage devices. The next level of abstraction is volume groups which can be created out of multiple physical volumes. VGs are virtualized devices created from combining multiple physical volumes. A good analogy to describe the relationship between physical volumes and volume groups is that physical volumes are used to build volume groups. The final form of abstraction used is logical volumes which are what the data is actually stored on. Logical volumes are virtualized partitions and so they must be formatted with a partitioning scheme and a filesystem.

Why three levels of abstraction? Well, the first level allows multiple storage devices to be used by LVM as virtual drives. Without virtualized physical volumes LVM would not even function. For the second level, volume groups are necessary since they are what allow LVM to abstract multiple devices as one drive. Then logical volumes are needed to allow users to create partitions off of the virtual device created from the PVs.

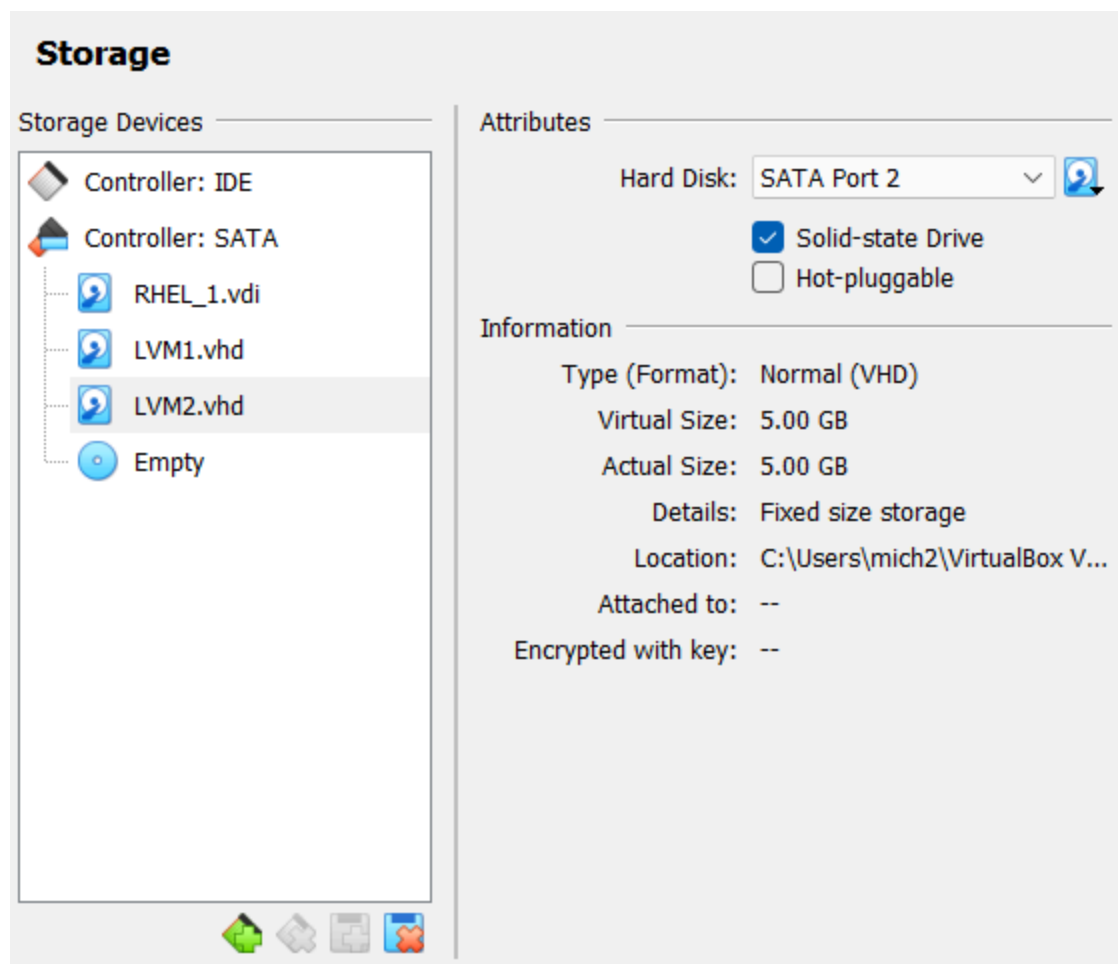
1.4 Create two 5 GB VHDs on Virtualbox:

Both drives will be configured the same way.





1.5 Attach the drives to the Linux VM and verify they are detected by it.



The two 5 GB LVM drives are attached to my Linux VM. Now I will verify that they are detected by the VM.

```
[administrator@rhel-server1 ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                7:0      0  74.2M  1 loop /var/lib/snapd/snap/core22/1380
loop1                7:1      0  38.8M  1 loop /var/lib/snapd/snap/snapd/21759
loop2                7:2      0  19.2M  1 loop /var/lib/snapd/snap/glances/1866
sda                  8:0      0   25G  0 disk
├─sda1                8:1      0    1G  0 part /boot
└─sda2                8:2      0   24G  0 part
   └─rhel-root         253:0      0  21.5G  0 lvm  /
      └─rhel-swap       253:1      0   2.5G  0 lvm  [SWAP]
sdb                  8:16     0    5G  0 disk
sdc                  8:32     0    5G  0 disk
sr0                 11:0     1 1024M  0 rom
[administrator@rhel-server1 ~]$
```

The new 5 GB drives are sdb and sdc.

Section #2 Set up Physical Volumes:

2.1 Initialize both LVM devices as PVs

Note: I am not planning on partitioning these devices so I will not be using fdisk or gparted to put MBR or GPT on these drives.

To make drives work within LVM you first need to create physical volumes out of them using the PVcreate command.

```
[administrator@rhel-server1 ~]$ sudo pvcreate /dev/sdb /dev/sdc
Physical volume "/dev/sdb" successfully created.
Physical volume "/dev/sdc" successfully created.
[administrator@rhel-server1 ~]$ pvs
WARNING: Running as a non-root user. Functionality may be unavailable.
/run/lock/lvm/P_global:aux: open failed: Permission denied
[administrator@rhel-server1 ~]$ sudo pvs
PV          VG      Fmt  Attr PSize  PFree
/dev/sda2   rhel    lvm2 a--  <24.00g  0
/dev/sdb          lvm2 ---   5.00g  5.00g
/dev/sdc          lvm2 ---   5.00g  5.00g
[administrator@rhel-server1 ~]$
```

You can view physical volume information using the PVs command.

2.2 Use PVdisplay to view detailed information about the PVs

If you want more details on the PVs you can use PVdisplay.

```
[administrator@rhel-server1 ~]$ sudo pvdisplay /dev/sdb
"/dev/sdb" is a new physical volume of "5.00 GiB"
--- NEW Physical volume ---
PV Name           /dev/sdb
VG Name
PV Size           5.00 GiB
Allocatable       NO
PE Size           0
Total PE          0
Free PE           0
Allocated PE      0
PV UUID           C3vp5k-mybx-4g3W-S5Zv-HB0L-0IA4-h4qi2x
[administrator@rhel-server1 ~]$
```

```
[administrator@rhel-server1 ~]$ sudo pvdisplay /dev/sdc
"/dev/sdc" is a new physical volume of "5.00 GiB"
--- NEW Physical volume ---
PV Name               /dev/sdc
VG Name
PV Size               5.00 GiB
Allocatable           NO
PE Size               0
Total PE              0
Free PE               0
Allocated PE          0
PV UUID               zf8ow6-XzC1-gRZX-v3HS-3DMW-eNy0-LSKNdy

[administrator@rhel-server1 ~]$
```

Section #3 Create and Manage VGs:

3.1 Create a VG out of /dev/sdb

After physical volumes are created, volume groups can be created out of combining them.

```
[administrator@rhel-server1 ~]$ sudo vgcreate lvmtest /dev/sdb
[sudo] password for administrator:
Volume group "lvmtest" successfully created
```

Note that I only added /dev/sdb to the lvmtest VG.

3.2 View detailed information about this volume group


```
[administrator@rhel-server1 ~]$ sudo vgdisplay lvmtest
--- Volume group ---
VG Name                lvmtest
System ID
Format                 lvm2
Metadata Areas         1
Metadata Sequence No   1
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                 0
Max PV                  0
Cur PV                 1
Act PV                  1
VG Size                 <5.00 GiB
PE Size                 4.00 MiB
Total PE                1279
Alloc PE / Size         0 / 0
Free PE / Size          1279 / <5.00 GiB
VG UUID                 Gcx6ce-nPtI-tnaL-7Pxc-XFVQ-qipl-MZR1sW

[administrator@rhel-server1 ~]$
```

VG information can be found by using the `vgdisplay` command.

3.3 Add `/dev/sdc` to the volume group

```
[administrator@rhel-server1 ~]$ sudo vgextend lvmtest /dev/sdc
Volume group "lvmtest" successfully extended
[administrator@rhel-server1 ~]$
```

The size of a volume group can be increased by adding more physical volumes to it with the `vgextend` command.

3.4 Verify that the VG expanded

```
[administrator@rhel-server1 ~]$ sudo vgdisplay lvmtest
--- Volume group ---
VG Name                lvmtest
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   2
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                 0
Max PV                 0
Cur PV                 2
Act PV                 2
VG Size                 9.99 GiB
PE Size                 4.00 MiB
Total PE                2558
Alloc PE / Size        0 / 0
Free PE / Size          2558 / 9.99 GiB
VG UUID                 Gcx6ce-nPtI-tnaL-7Pxc-XFVQ-qipl-MZR1sW

[administrator@rhel-server1 ~]$
```

3.5 Remove /dev/sdc from the VG

```
[administrator@rhel-server1 ~]$ sudo vgreduce lvmtest /dev/sdc
Removed "/dev/sdc" from volume group "lvmtest"
[administrator@rhel-server1 ~]$ sudo vgdisplay
--- Volume group ---
VG Name                lvmtest
System ID
Format                 lvm2
Metadata Areas         1
Metadata Sequence No   3
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                 0
Max PV                 0
Cur PV                 1
Act PV                 1
VG Size                 <5.00 GiB
PE Size                 4.00 MiB
Total PE                1279
Alloc PE / Size        0 / 0
Free PE / Size          1279 / <5.00 GiB
VG UUID                 Gcx6ce-nPtI-tnaL-7Pxc-XFVQ-qipl-MZR1sW
```

If you want to shrink a VG you would need to use the `vgreduce` command to remove a PV from it.

Section #4 Create and Manage LVs:

Creating logical volumes can be done with around 3 steps depending on if you want to partition the logical volume or not. I am not partitioning the storage devices (I would have done that earlier) so I can go straight into formatting the LV with a filesystem.

4.1 Configure the Logical Volume:

I will add the /dev/sdc drive back the the VG lvmtest.

```
[administrator@rhel-server1 ~]$ sudo vgextend lvmtest /dev/sdc
Volume group "lvmtest" successfully extended
[administrator@rhel-server1 ~]$ sudo vgdisplay
--- Volume group ---
VG Name                lvmtest
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   4
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                 0
Max PV                 0
Cur PV                 2
Act PV                 2
VG Size                 9.99 GiB
PE Size                 4.00 MiB
Total PE                2558
Alloc PE / Size         0 / 0
Free PE / Size          2558 / 9.99 GiB
VG UUID                 Gcx6ce-nPtI-tnaL-7Pxc-XFVQ-qipL-MZR1sW
```

Now that I have both disks in the VG pool I will carve out a 8 GB LV from the VG lvmtest.

```
[administrator@rhel-server1 ~]$ sudo lvcreate -L 8GB -n testlv lvmtest
Logical volume "testlv" created.
[administrator@rhel-server1 ~]$
```

```

[administrator@rhel-server1 ~]$ sudo lvdisplay
--- Logical volume ---
LV Path                /dev/lvmtest/testlv
LV Name                 testlv
VG Name                lvmtest
LV UUID                hSDotj-pma6-k94E-0aJT-MRPs-M3C0-hhwYcA
LV Write Access         read/write
LV Creation host, time rhel-server1, 2025-03-19 13:04:18 -0700
LV Status               available
# open                  0
LV Size                 8.00 GiB
Current LE              2048
Segments                2
Allocation              inherit
Read ahead sectors      auto
- currently set to      256
Block device            253:2

```

LV information can be displayed using the `lvdisplay` command.

4.2 Apply the filesystem to the logical volume:

Once the LV is created a filesystem can be applied to it. I am using ext4.

```

[administrator@rhel-server1 ~]$ sudo mkfs.ext4 /dev/lvmtest/testlv
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2097152 4k blocks and 524288 inodes
Filesystem UUID: 47de89f1-f90a-4d6a-9d22-e9372bbad49a
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

```

```

Disk /dev/mapper/lvmtest-testlv: 8 GiB, 8589934592 bytes, 16777216 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

```

Section #5 Resize Logical Volume:

The size of a logical volume can be reduced or increased using the `lvextend` and `lvreduce` commands. Since the LV size will increase or decrease you need to resize the filesystem format

appropriately. Luckily I haven't mounted my logical volume yet so I can proceed with resizing this LV without my data being affected. **You need to unmount the lv to resize!**

5.1 Reduce the size of the LV to 5GB, resizing the filesystem after:

```
[administrator@rhel-server1 mapper]$ sudo lvreduce --resizefs -L 5GB lvmtest/testlv
File system ext4 found on lvmtest/testlv.
File system size (8.00 GiB) is larger than the requested size (5.00 GiB).
File system reduce is required using resize2fs.
File system fsck will be run before reduce.
Reducing file system ext4 to 5.00 GiB (5368709120 bytes) on lvmtest/testlv...
e2fsck /dev/lvmtest/testlv
/dev/lvmtest/testlv: 11/524288 files (0.0% non-contiguous), 58511/2097152 blocks
e2fsck done
resize2fs /dev/lvmtest/testlv 5242880k
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/lvmtest/testlv to 1310720 (4k) blocks.
The filesystem on /dev/lvmtest/testlv is now 1310720 (4k) blocks long.

resize2fs done
Reduced file system ext4 on lvmtest/testlv.
Size of logical volume lvmtest/testlv changed from 8.00 GiB (2048 extents) to 5.00 GiB (1280 extents).
Logical volume lvmtest/testlv successfully resized.
[administrator@rhel-server1 mapper]$
```

I used the resizefs flag to automatically resize the filesystem while the LV is shrunk.

5.2 Increase the size of the LV back to 8 GB, resizing the filesystem after:

```
[administrator@rhel-server1 mapper]$ sudo lvextend --resizefs -L 8GB lvmtest/testlv
Size of logical volume lvmtest/testlv changed from 5.00 GiB (1280 extents) to 8.00 GiB (2048 extents).
File system ext4 found on lvmtest/testlv.
File system fsck will be run before extend.
Extending file system ext4 to 8.00 GiB (8589934592 bytes) on lvmtest/testlv...
e2fsck /dev/lvmtest/testlv
/dev/lvmtest/testlv: 11/327680 files (0.0% non-contiguous), 45150/1310720 blocks
e2fsck done
resize2fs /dev/lvmtest/testlv
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/lvmtest/testlv to 2097152 (4k) blocks.
The filesystem on /dev/lvmtest/testlv is now 2097152 (4k) blocks long.

resize2fs done
Extended file system ext4 on lvmtest/testlv.
Logical volume lvmtest/testlv successfully resized.
[administrator@rhel-server1 mapper]$
```

Section #6 Use LVM Mirroring:

Since I combined my current PVs into a VG I can't use them for mirroring. I will need to create a new virtual storage device, configure it as a PV, and add this new PV to my existing VG (lvmtest). The reason why is that my VG does not have sufficient storage to store the mirrored data.

6.1 Create the new storage device:

The screenshot shows the 'Create New Virtual Disk' dialog box. The 'Type' is set to 'Normal'. The 'Location' is 'E:\mich2\mich2\IT Labs\VMs\VM Images\RHEL_1\RHEL_1\NewVirtualDisk.vhd'. The 'Description' field is empty. The 'Size' is set to 10.00 GB, with a slider bar showing the range from 4.00 MB to 2.00 TB. The 'Apply' and 'Reset' buttons are at the bottom right.

6.2. Attach the new drive

The screenshot shows the 'Storage' settings window. The 'Storage Devices' list on the left includes 'Controller: IDE', 'Controller: SAT...', 'RHEL_1.vdi', 'LVM1.vhd', 'LVM2.vhd', 'Empty', and 'NewVirtualDisk.vhd'. The 'Attributes' section on the right shows 'Name: SATA', 'Type: AHCI', 'Port Count: 5', and 'Use Host I/O Cache' checked. The 'OK', 'Cancel', and 'Help' buttons are at the bottom.

6.3 Format the drive as a PV:

```
[administrator@rhel-server1 ~]$ lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0  74.2M  1 loop /var/lib/snapd/snap/core22/1380
loop1                 7:1      0  38.8M  1 loop /var/lib/snapd/snap/snapd/21759
loop2                 7:2      0  19.2M  1 loop /var/lib/snapd/snap/glances/1866
sda                   8:0       0    5G   0 disk
└─lvmtest-testlv     253:2     0    8G   0 lvm
sdb                   8:16      0   10G   0 disk
sdc                   8:32      0    5G   0 disk
└─lvmtest-testlv     253:2     0    8G   0 lvm
sdd                   8:48      0   25G   0 disk
├─sdd1                8:49      0    1G   0 part /boot
└─sdd2                8:50      0   24G   0 part
   ├─rhel-root         253:0     0  21.5G   0 lvm /
   └─rhel-swap         253:1     0   2.5G   0 lvm [SWAP]
sr0                   11:0      1 1024M   0 rom
[administrator@rhel-server1 ~]$
```

The drive is sdb.

```
[administrator@rhel-server1 ~]$ sudo pvcreate /dev/sdb
[sudo] password for administrator:
Physical volume "/dev/sdb" successfully created.
[administrator@rhel-server1 ~]$
```

I turned /dev/sdb into a PV.

6.4 Add the PV to the existing VG:

```
[administrator@rhel-server1 ~]$ sudo vgextend lvmtest /dev/sdb
Volume group "lvmtest" successfully extended
```

```

--- Volume group ---
VG Name          lvmtest
System ID
Format           lvm2
Metadata Areas   3
Metadata Sequence No 8
VG Access        read/write
VG Status        resizable
MAX LV           0
Cur LV          1
Open LV          0
Max PV           0
Cur PV          3
Act PV           3
VG Size          <19.99 GiB
PE Size          4.00 MiB
Total PE         5117
Alloc PE / Size  2048 / 8.00 GiB
Free PE / Size   3069 / <11.99 GiB
VG UUID          Gcx6ce-nPtI-tnaL-7Pxc-XFVQ-qipL-MZR1sW

```

Now the volume group has 19.99 GB (twice what it had before).

6.5 Convert the LV to a RAID 1 array using itself and the new 10 GB added to the VG:

```

[administrator@rhel-server1 ~]$ sudo lvconvert --type raid1 --mirrors 1 lvmtest/testlv
Are you sure you want to convert linear LV lvmtest/testlv to raid1 with 2 images enhancing resilience? [y/n]: y
Logical volume lvmtest/testlv successfully converted.
[administrator@rhel-server1 ~]$

```

I used lvconvert to convert the logical volume into a RAID 1 array.

```

[administrator@rhel-server1 ~]$ sudo lvs
LV      VG      Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
testlv  lvmtest rwi-a-r--- 8.00g                                     74.62
root    rhel     -wi-ao---- <21.50g
swap    rhel     -wi-ao---- 2.50g
[administrator@rhel-server1 ~]$

```

Testlv is now a RAID 1 array since data is being synced as you can see in the lvs command output.


```
[administrator@rhel-server1 ~]$ sudo lvdisplay -m /dev/lvmtest/testlv
[sudo] password for administrator:
--- Logical volume ---
LV Path                /dev/lvmtest/testlv
LV Name                 testlv
VG Name                 lvmtest
LV UUID                 hSDotj-pma6-k94E-0aJT-MRPs-M3C0-hhwYcA
LV Write Access         read/write
LV Creation host, time rhel-server1, 2025-03-19 13:04:18 -0700
LV Status                available
# open                  0
LV Size                 8.00 GiB
Current LE              2048
Mirrored volumes        2
Segments                1
Allocation              inherit
Read ahead sectors      auto
- currently set to      256
Block device            253:2

--- Segments ---
Logical extents 0 to 2047:
  Type                  raid1
  Monitoring             monitored
  Raid Data LV 0
    Logical volume       testlv_rimage_0
    Logical extents      0 to 2047
  Raid Data LV 1
    Logical volume       testlv_rimage_1
    Logical extents      0 to 2047
  Raid Metadata LV 0     testlv_rmeta_0
  Raid Metadata LV 1     testlv_rmeta_1
```

Using `lvdisplay` you can see even more information about the `testlv` logical volume.

Section #7 Mount LVM Persistently and Test It:

7.1 Create a mountpoint under the administrator home directory:

```
[administrator@rhel-server1 ~]$ mkdir LVM
[administrator@rhel-server1 ~]$ ls -l
total 0
drwxr-x---. 3 administrator administrator 30 Feb  2 16:00 Desktop
drwxr-x---. 3 administrator administrator 32 Aug  8 2024 Documents
drwxr-x---. 2 administrator administrator 56 Jul  7 2024 Downloads
drwxr-xr-x. 2 administrator administrator  6 Mar 20 18:29 LVM
drwxr-x---. 2 administrator administrator  6 Apr  4 2024 Music
drwxr-x---. 2 administrator administrator 52 Apr 19 2024 Pictures
drwxr-x---. 2 administrator administrator  6 Apr  4 2024 Public
drwxr-x---. 2 administrator administrator  6 Apr  4 2024 Templates
drwxr-x---. 2 administrator administrator  6 Apr  4 2024 Videos
drwxr-x---. 3 administrator web_admin      43 Aug 16 2024 web_content
[administrator@rhel-server1 ~]$
```

The LVM logical volume will be mounted on the administrator user's home directory.

7.2 Automount the LVM LV:

```
GNU nano 5.6.1 /etc/fstab

#
# /etc/fstab
# Created by anaconda on Fri Apr  5 00:16:35 2024
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rhel-root / xfs defaults 0 0
UUID=55b00bfb-b167-4acc-b77c-0692a7385352 /boot xfs defaults 0 0
/dev/mapper/rhel-swap none swap defaults 0 0

#Mounting LVM
/dev/lvmtest/testlv /home/administrator/LVM ext4 defaults 0 0
```

To automount the LV you put the device information into /etc/fstab like you would do for my other storage device. Note that I did not use the UUID for the device, only the mapper name. Still, I had to include the mount point, the filesystem type, and the other settings.

7.3 Create a file in the mount point directory:

```
administrator@rhel-server1:~/LVM — nano test1.txt

GNU nano 5.6.1 test1.txt
Hello! Testing testing 123!! Testing LVM!
```

7.4 Download a file from the internet into this directory as well:

```
[administrator@rhel-server1 LVM]$ ls -l
total 1092
-rw-r--r--. 1 administrator administrator 1093835 Mar 20 19:06 Apple_10K.pdf
drwx-----. 2 root          root          16384 Mar 19 13:06 lost+found
-rw-r--r--. 1 administrator administrator    43 Mar 20 18:59 test1.txt
[administrator@rhel-server1 LVM]$
```

I downloaded an investor presentation from a publicly traded company.

7.5 Reboot system and check if the LV is mounted:

```
[administrator@rhel-server1 ~]$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                              7:0      0 38.8M  1 loop /var/lib/snapd/snap/snapd/21759
loop1                              7:1      0 19.2M  1 loop /var/lib/snapd/snap/glances/1866
loop2                              7:2      0 74.2M  1 loop /var/lib/snapd/snap/core22/1380
sda                                8:0      0   5G   0 disk
├─lvmtest-testlv_rimage_0
│   └─lvmtest-testlv          253:6    0   8G   0 lvm  /home/administrator/LVM
sdb                                8:16     0   5G   0 disk
├─lvmtest-testlv_rmeta_0
│   └─lvmtest-testlv          253:6    0   8G   0 lvm  /home/administrator/LVM
└─lvmtest-testlv_rimage_0
    └─lvmtest-testlv          253:6    0   8G   0 lvm  /home/administrator/LVM
sdc                                8:32     0  10G   0 disk
├─lvmtest-testlv_rmeta_1
│   └─lvmtest-testlv          253:6    0   8G   0 lvm  /home/administrator/LVM
└─lvmtest-testlv_rimage_1
    └─lvmtest-testlv          253:6    0   8G   0 lvm  /home/administrator/LVM
sdd                                8:48     0  25G   0 disk
├─sdd1                            8:49     0   1G   0 part /boot
└─sdd2                            8:50     0  24G   0 part
    ├─rhel-root                    253:0    0 21.5G   0 lvm  /
    └─rhel-swap                    253:1    0   2.5G   0 lvm  [SWAP]
sr0                                11:0     1 1024M   0 rom
```

7.6 Test what happens if a drive is removed:

```
[administrator@rhel-server1 ~]$ md5sum LVM/test1.txt
77a6e29ea9d52d365a9a5afae5b5164a  LVM/test1.txt
[administrator@rhel-server1 ~]$
```

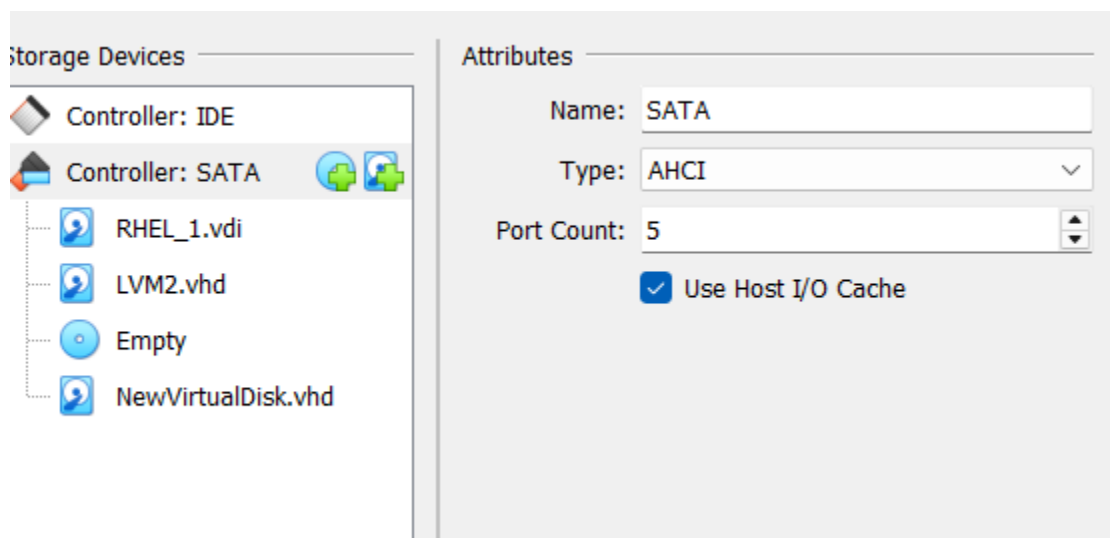
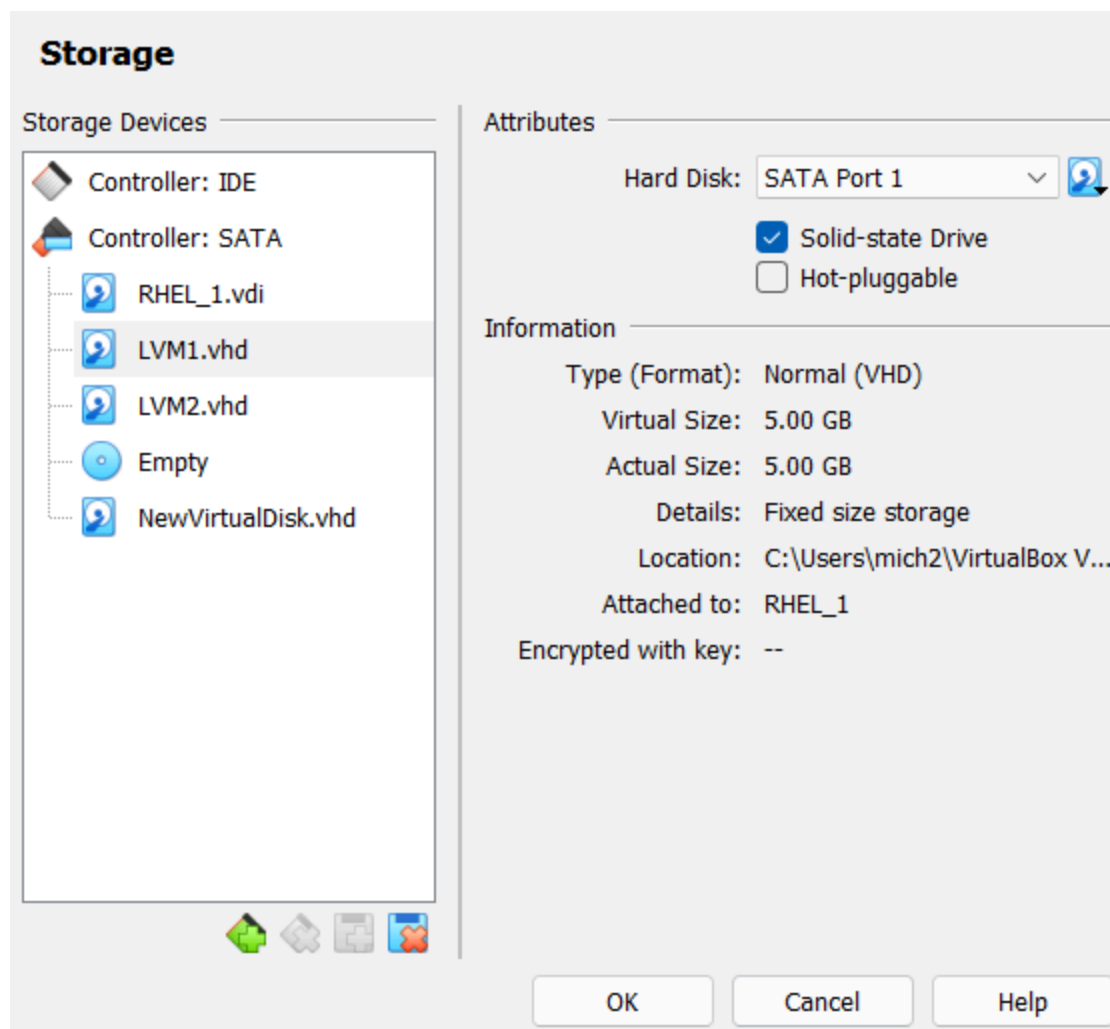
I created a hash of the file so I can verify that the data is the same after I remove a drive from the RAID array.

```
administrator@rhel-server1:~ — udevadm info --query=all --name=/dev/sdc
```

```
S: disk/by-id/scsi-0ATA_VBOX_HARDDISK_VB9770bca6-1c916284  
S: disk/by-id/lvm-pv-uuid-zf8ow6-XzC1-gRZX-v3HS-3DMW-eNy0-LSKNdy  
S: disk/by-id/scsi-1ATA_VBOX_HARDDISK_VB9770bca6-1c916284  
S: disk/by-id/scsi-SATA_VBOX_HARDDISK_VB9770bca6-1c916284  
S: disk/by-path/pci-0000:00:0d.0-ata-3  
S: disk/by-path/pci-0000:00:0d.0-ata-3.0  
S: disk/by-id/ata-VBOX_HARDDISK_VB9770bca6-1c916284  
S: disk/by-diskseq/3  
Q: 3  
E: DEVPATH=/devices/pci0000:00/0000:00:0d.0/ata5/host4/target4:0:0/4:0:0:0/block/sdc  
E: DEVNAME=/dev/sdc  
E: DEVTYPEDISK  
E: DISKSEQ=3  
E: MAJOR=8  
E: MINOR=32  
E: SUBSYSTEM=block  
E: USEC_INITIALIZED=14969834  
E: ID_ATA=1  
E: ID_TYPE=disk  
E: ID_BUS=ata  
E: ID_MODEL=VBOX_HARDDISK  
E: ID_MODEL_ENC=VBOX\x20HARDDISK\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20\x20  
E: ID_REVISION=1.0
```

lines 8-30/68 42%

Remove /dev/sdc which is on Sata port 1.



Now LVM1 is removed! I then booted up the VM once again and ran into a boot issue.

```

Starting NFS status monitor for NFSv2/3 locking....
[ OK ] Started NFS Mount Daemon.
[ OK ] Started NFS status monitor for NFSv2/3 locking..
Starting NFS server and services...
[ OK ] Started Security Auditing Service.
Starting Record System Boot/Shutdown in UTMP...
[ OK ] Finished Record System Boot/Shutdown in UTMP.
Starting Record Runlevel Change in UTMP...
[ OK ] Finished Record Runlevel Change in UTMP.
[ OK ] Finished NFS server and services.
Starting Notify NFS peers of a restart...
[ OK ] Started Notify NFS peers of a restart.
[ OK ] Finished Crash recovery kernel arming.
You are in emergency mode. After logging in, type "journalctl -xb" to view
system logs, "systemctl reboot" to reboot, "systemctl default" or "exit"
to boot into default mode.
Give root password for maintenance
(or press Control-D to continue):

```

Not surprised the boot process had an issue. I will remove the PV from the vg. It thinks there is an extra PV for the lvmtest VG but obviously /dev/sdc was removed, causing the boot process to be interrupted and stopped. Basically the kernel is saying “hey, where is /dev/sdc? It should be attached to the system to be used in the lvmtest VG!” It is an easy fix.

```

Starting Notify NFS peers of a restart...
[ OK ] Started Notify NFS peers of a restart.
You are in emergency mode. After logging in, type "journalctl -xb" to view
system logs, "systemctl reboot" to reboot, "systemctl default" or "exit"
to boot into default mode.
Give root password for maintenance
(or press Control-D to continue):
Login incorrect

Give root password for maintenance
(or press Control-D to continue):
[root@rhel-server1 ~]#

```

I first entered recovery mode in the boot menu. I then had to use vgreduce to remove the missing drive (/dev/sdb).

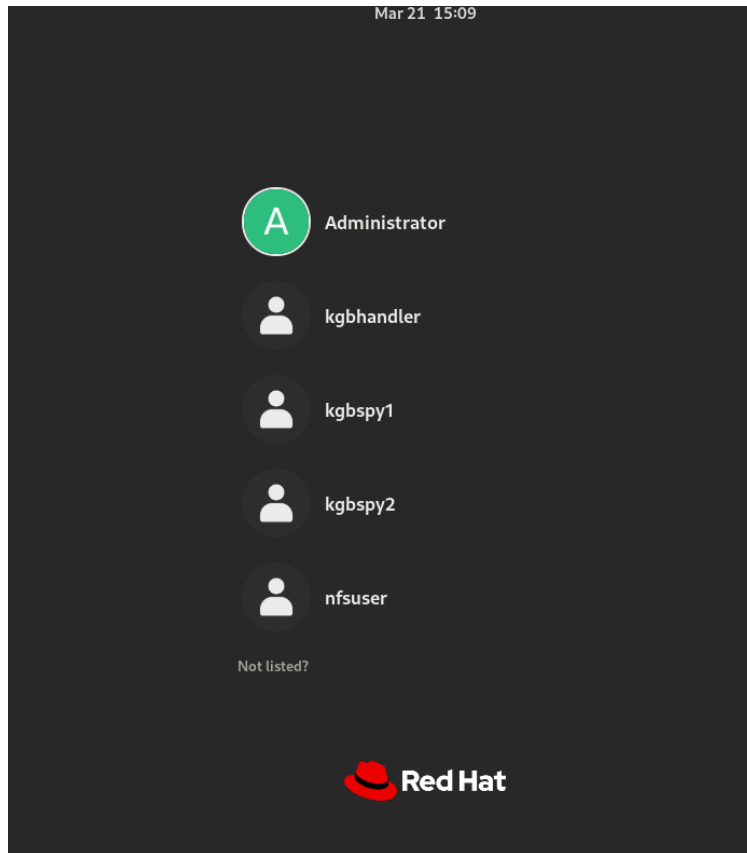
```

[root@rhel-server1 ~]# vgreduce --removemissing --mirroronly --force lvmtest
WARNING: Couldn't find device with uuid C3vp5k-mybx-4g3W-S5Zv-HB0L-0IA4-h4qi2x.
WARNING: VG lvmtest is missing PV C3vp5k-mybx-4g3W-S5Zv-HB0L-0IA4-h4qi2x (last written to [unknown])
WARNING: Couldn't find device with uuid C3vp5k-mybx-4g3W-S5Zv-HB0L-0IA4-h4qi2x.
WARNING: Couldn't find device with uuid C3vp5k-mybx-4g3W-S5Zv-HB0L-0IA4-h4qi2x.
Wrote out consistent volume group lvmtest.
[root@rhel-server1 ~]# pvs
PV          VG      Fmt Attr PSize  PFree
/dev/sda2   rhel    lvm2 a--  <24.00g  0
/dev/sdb    lvmtest lvm2 a--  <5.00g  <5.00g
/dev/sdc    lvmtest lvm2 a--  <10.00g  1.99g
[root@rhel-server1 ~]# pvdisplay
--- Physical volume ---
PV Name           /dev/sdb
VG Name           lvmtest
PV Size           5.00 GiB / not usable 4.00 MiB
Allocatable       yes
PE Size           4.00 MiB
Total PE          1279
Free PE           1279
Allocated PE      0
PV UUID           zf8ow6-XzC1-gRZX-v3HS-3DMW-eNy0-LSKNdy

--- Physical volume ---
PV Name           /dev/sdc
VG Name           lvmtest
PV Size           10.00 GiB / not usable 4.00 MiB
Allocatable       yes
PE Size           4.00 MiB
Total PE          2559
Free PE           510
Allocated PE      2049
PV UUID           gYPa5r-UIuP-Kq0n-y1tU-4P5l-lxgS-w0mfHd

--- Physical volume ---
PV Name           /dev/sda2
VG Name           rhel
PV Size           <24.00 GiB / not usable 3.00 MiB
Allocatable       yes (but full)
PE Size           4.00 MiB
Total PE          6143
Free PE           0
Allocated PE      6143
PV UUID           Whd7m1-H0Ir-e3r1-5cFt-HUdT-nU7R-QoTzE0
[root@rhel-server1 ~]#

```



```
[administrator@rhel-server1 LVM]$ sudo lvs
WARNING: RaidLV lvmtest/testlv needs to be refreshed! See character 'r' at position 9
attributes and also its SubLV(s) with option '-a'.
LV      VG      Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
testlv  lvmtest rwi-aor-r- 8.00g
root    rhel    -wi-ao---- <21.50g
swap    rhel    -wi-ao---- 2.50g
[administrator@rhel-server1 LVM]$
```

The RAID array has to be refreshed since there is no extra storage in the VG. Because of that the mirroring can't occur and the system is not happy. I am okay with leaving it since my data has been saved at least. The RAID 1 array could be fixed if I added another PV but for the purposes of this lab I am fine. The hash is the same post-removal so that means the data was saved!

```
[administrator@rhel-server1 LVM]$ ls -l
total 1092
-rw-r--r--. 1 administrator administrator 1093835 Mar 20 19:06 Apple_10K.pdf
drwx-----. 2 root                root          16384 Mar 19 13:06 lost+found
-rw-r--r--. 1 administrator administrator    43 Mar 20 18:59 test1.txt
[administrator@rhel-server1 LVM]$ md5sum test1.txt
77a6e29ea9d52d365a9a5afaeeb5164a test1.txt
[administrator@rhel-server1 LVM]$
```